

1. To prove $A \subseteq B \Leftrightarrow A \cap B^c = \emptyset$, prove in both directions:

1.1) $A \subseteq B \Rightarrow A \cap B^c = \emptyset$

Let $x \in A$

by assumption

$x \in B$

by subset law

$x \notin B^c$

by complement law

$x \notin A \cap B^c$

by the above statement

$\therefore A \cap B^c = \emptyset$

1.2) $A \cap B^c = \emptyset \Rightarrow A \subseteq B$

Let $x \in A$

by assumption

$x \notin B^c$

by intersection

$x \in B$

by complement law

$x \in A \rightarrow x \in B$

by the above statements

$\therefore A \subseteq B$

To prove $A \subseteq B \Leftrightarrow A \cup B = B$, prove in both directions:

2.1) $A \subseteq B \Rightarrow A \cup B = B$

Let $x \in A$

by assumption

$x \in B$

by subset

$x \in A \vee x \in B$

by above statements

$((x \in A) \vee (x \in B)) = ((x \in B) \vee (x \in B))$

because everything in A is in B

$A \cup B = B \cup B$

$B \cup B = B$

by idempotent law

$\therefore A \cup B = B$

2.2) $A \cup B = B \Rightarrow A \subseteq B$

Let $x \in A$

by assumption

$x \in A \cup B$

by union

$x \in B$

by premise.

$x \in A \rightarrow x \in B$

by above statements

$\therefore A \subseteq B$

2. To prove $A \setminus (B \setminus C) = B \setminus (C \setminus A) \Leftrightarrow A \subseteq B$ and $B \subseteq C$, prove in both directions:

2.1) $A \setminus (B \setminus C) = B \setminus (C \setminus A) \Rightarrow A \subseteq B$ and $B \subseteq C$

$A \setminus (B \setminus C) = B \setminus (C \setminus A)$
 $A \cap (B \setminus C)^c = B \cap (C \setminus A)^c$ by difference law
 $A \cap (B \cap C^c)^c = B \cap (C \cap A^c)^c$ by difference law
 $A \cap (B^c \cup C) = B \cap (C^c \cup A)$ by De Morgan's law
 $(A \cap B^c) \cup (A \cap C) = (B \cap C^c) \cup (B \cap A)$ by Distributive laws
 Let $x \in ((A \cap B^c) \cup (A \cap C)) = \text{Let } x \in ((B \cap C^c) \cup (B \cap A))$ by assumption
 $(x \in A \cap x \notin B) \cup (x \in A \cap x \in C) = (x \in B \cap x \notin C) \cup (x \in B \cap x \in A)$ by intersection
 $x \in A \cup (x \in A \cap x \in C) = x \in B \cup (x \in B \cap x \in A)$ by absorption law
 $A \setminus (B \setminus C) = A$ and $B \setminus (C \setminus A) = B$ by absorption law
 $\therefore A \setminus (B \setminus C) = B \setminus (C \setminus A) \Rightarrow A \subseteq B$ and $B \subseteq C$

■

3. To prove $A \setminus (B \setminus C) = B \setminus (C \setminus A) = C \setminus (A \setminus B) \Leftrightarrow A = B = C$, prove in both directions:

3.1) $A \setminus (B \setminus C) = B \setminus (C \setminus A) = C \setminus (A \setminus B) \Rightarrow A = B = C$

Let $x \in A \setminus (B \setminus C)$ by assumption
 $x \in A$ and $x \notin (B \setminus C)$ by difference logic
 $\therefore A \setminus (B \setminus C) = A$

If $A \setminus (B \setminus C) = B \setminus (C \setminus A)$
 Let $x \in B \setminus (C \setminus A)$ by assumption
 $x \in B$ and $x \notin (C \setminus A)$ by difference logic
 $\therefore B \setminus (C \setminus A) = B$

If $B \setminus (C \setminus A) = C \setminus (A \setminus B)$
 Let $x \in C \setminus (A \setminus B)$ by assumption
 $x \in C$ and $x \notin (A \setminus B)$ by difference logic
 $\therefore C \setminus (A \setminus B) = C$

$\therefore A \setminus (B \setminus C) = B \setminus (C \setminus A) = C \setminus (A \setminus B) \Rightarrow A = B = C$

3.2) $A = B = C \Rightarrow A \setminus (B \setminus C) = B \setminus (C \setminus A) = C \setminus (A \setminus B)$

Let $x \in A$ by assumption
 Then $x \in B$ and $x \in C$ by equality logic
 Thus $B \setminus C$ is $\emptyset$ by rules of equality
 $\therefore A \setminus (B \setminus C) = A$ by difference law

Let $x \in A$ and $x \in B$ and $x \in C$ by assumption
 Thus $C \setminus A$ is $\emptyset$ by rules of equality
 $\therefore B \setminus (C \setminus A) = B$ by difference law

Let $x \in A$ and $x \in B$ and $x \in C$ by assumption
 Thus $A \setminus B$ is $\emptyset$ by rules of equality
 $\therefore C \setminus (A \setminus B) = C$ by difference law

$$\therefore A = B = C \Rightarrow A \setminus (B \setminus C) = B \setminus (C \setminus A) = C \setminus (A \setminus B)$$

